SynapseGrid

Verified renewable-energy dispatch for critical community loads



Prototype dashboard (simulated data)

Google Cloud AI Builder Cup · Category: Sustainability

Gemma reads the context · Google OR-Tools plans · an independent verifier decides

Solar can vanish in minutes

Island and remote communities are replacing expensive, polluting diesel with solar and batteries. That makes their small grids cleaner — and more volatile.



Sudden solar loss

A cloud front can remove most of a solar farm's output within minutes.



Unsafe recovery plans

A rushed plan can overload a power line, exceed a generator's rating, or drain a battery too far.



Critical loads at risk

The clinic, water pumps and emergency radio must stay powered, whatever else is cut.



No safety check

Small operators rarely have a tool that checks a plan against physical limits before acting on it.

Three roles that cannot overrule each other

Planners — including AI — may propose. Nothing is dispatched unless an independent check of every modeled physical limit passes.



Understand

Gemma

Reads the operator's free-text incident note and returns load priorities as JSON. Output is schema-validated and treated as untrusted.



Plan

Google OR-Tools

Finds the lowest-cost, lowest-carbon dispatch that balances power at every bus and respects line, battery and generator limits. Critical loads always served.

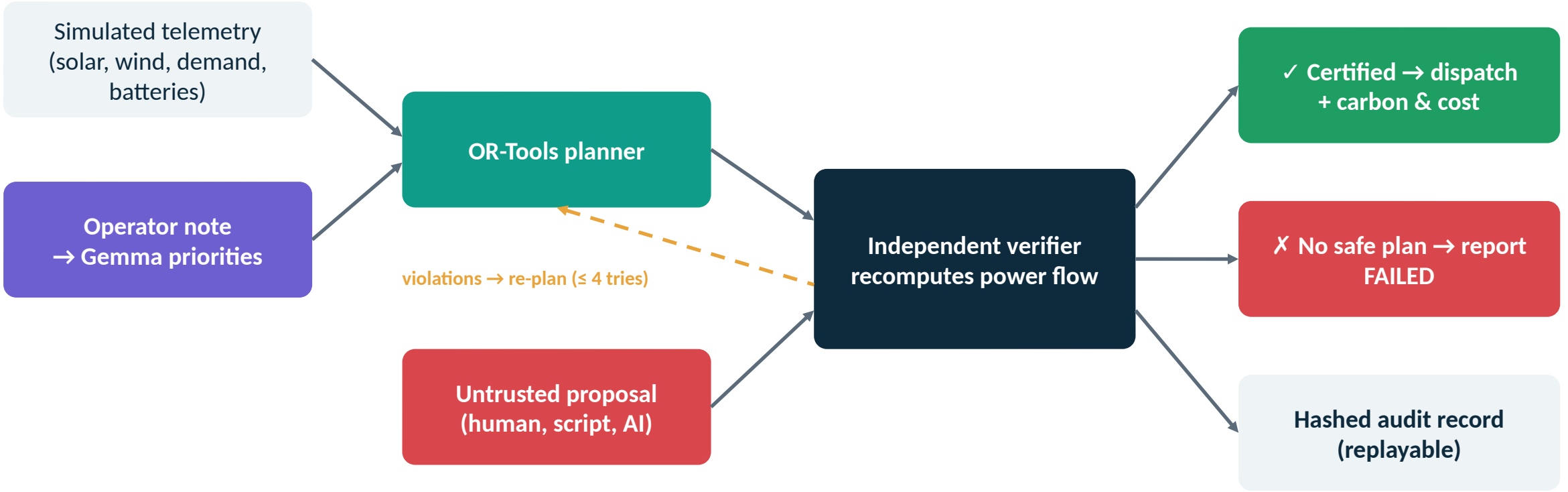


Decide

Independent verifier

Recomputes the physics from scratch and approves or rejects every proposal. Rejections go back to the planner with exact violations (max 4 tries).

How one recovery works



Why the verifier is independent: no shared code with the planner, Python standard library only, ignores the planner's claimed line flows, rejects unexpected fields, and recalculates every flow itself.

Demo: unsafe plan rejected, safe plan certified



INCIDENT RECOVERY · rejected vs certified dispatch

Gemma briefing

A1 REJECTED · 3 violation(s) → counter-example → planner cuts → A2 CERTIFIED · 27/27 checks

recovery 1.13 ms

✗ GEN_LIMIT[DG-HARB]: observed 2699 kW, expected ≤ 2000 kW (excess 698.6 kW) [major]

✗ GEN_RAMP[DG-HARB]: observed 2699 kW, expected ≤ 600 kW (excess 2099 kW) [major]

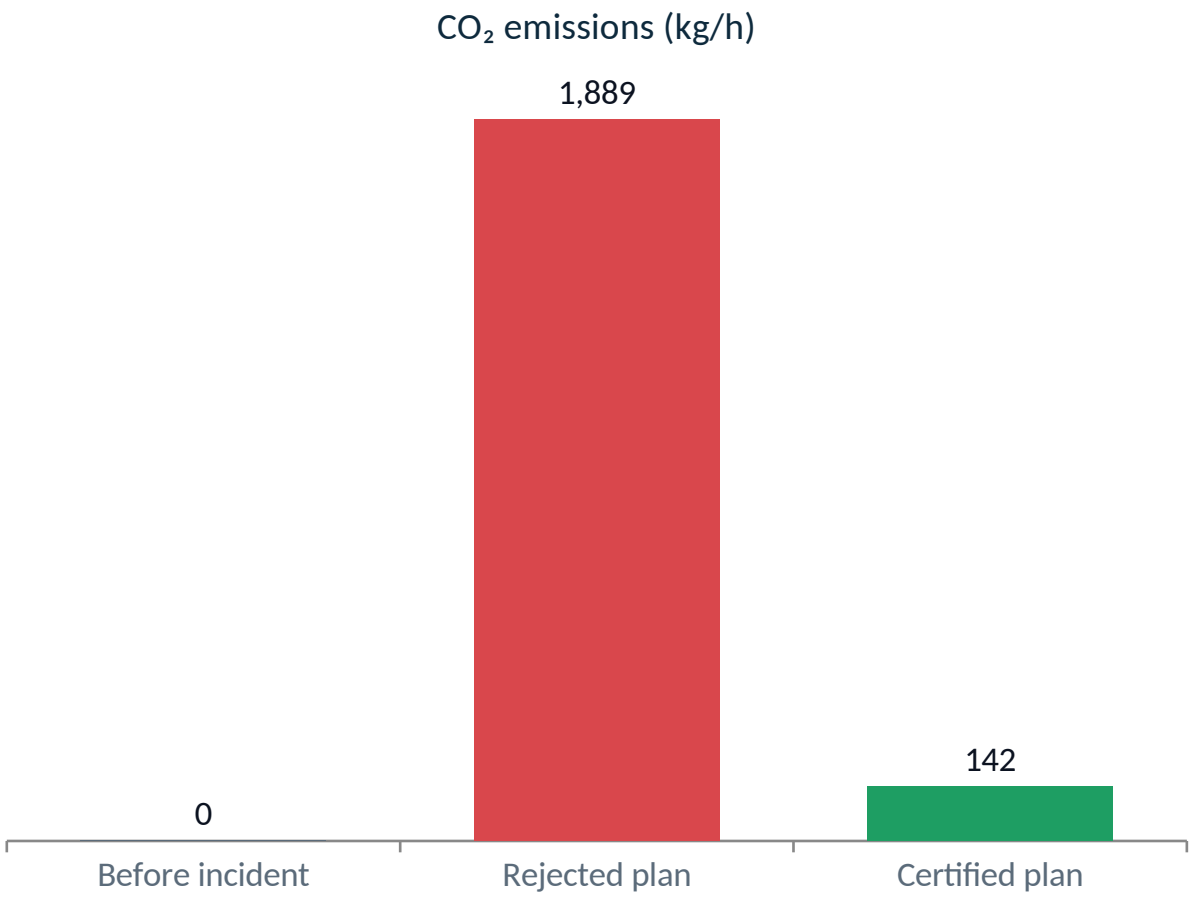
✗ LINE_CAPACITY[L15]: observed 1278 kW, expected ≤ 800 kW (excess 477.8 kW) [major]

METRIC	BEFORE INCIDENT certified baseline	REJECTED · A1 injected unsafe — NOT dispatched	CERTIFIED · A2 optimizer · dispatched
Emissions ($kgCO_2/h$)	0	1,889	142
Operating cost ($/h)	1	873	86
Critical-load coverage	100.0 %	100.0 %	100.0 %
Unmet demand (kW)	0	0	0
Renewable share	96.2 %	35.1 %	35.0 %
Diesel (kW)	0	2,699	203
Battery net (kW, + discharge)	115	-750	1,748

Deterministic incident: PV lost 1,836 kW (−75 %). Diesel 0.70 $kgCO_2/kWh$, renewables and battery 0 operational (declared assumptions). Priorities: default.

1. Cloud front removes 1,836 kW of solar (−75 %)
2. Proposal: restart diesel at 2,699 kW and charge the port battery
3. Rejected — 3 exact violations: generator rating, ramp limit, line L15 at 1,278 of 800 kW
4. OR-Tools re-plans with the violations as cuts
5. Certified — 27 / 27 checks pass; audit record replays to the same verdict

Cleaner, cheaper, critical loads kept on



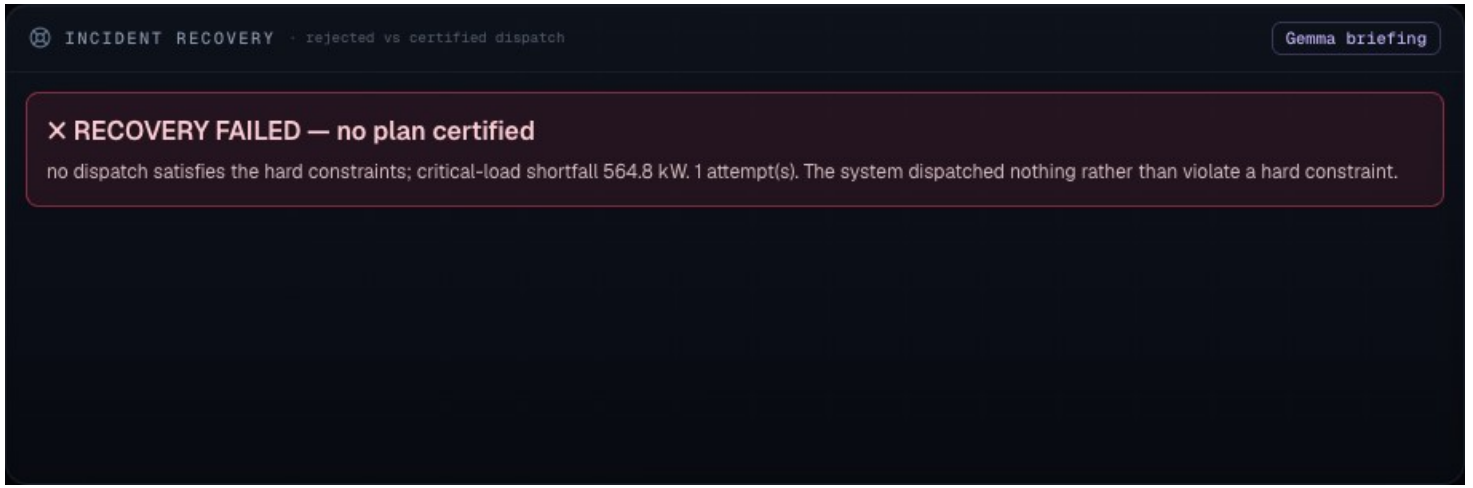
−1,747
kg CO_2/h
certified vs rejected plan

$86/h
vs $873/h
operating cost, certified vs rejected

100 %
critical loads
clinic, water and radio stay powered

Simulated data. Assumptions: diesel 0.70 kg CO_2/kWh and $320/MWh; solar, wind and batteries 0 operational emissions. Comparison is against the rejected plan, not a real-world baseline.

When no safe plan exists, it says so



Stress test

Night, no wind, batteries at 11 %, diesel genset tripped.

564.8 kW

critical-load shortfall reported — nothing dispatched

Why this matters

Critical loads are a hard constraint. The optimizer cannot trade them for cost or carbon, so when they can't be met the problem is infeasible and the system reports failure instead of shedding the clinic or pretending.

Evidence: 300 randomized fault scenarios

273 / 273

recoverable scenarios reached a certified plan after an unsafe proposal was injected

27 / 27

impossible scenarios reported as FAILED — never certified

0

certified plans found unsafe by a second, separately written checker

2,400 / 2,400

adversarial and malformed proposals classified the same by both checkers

1.1 ms

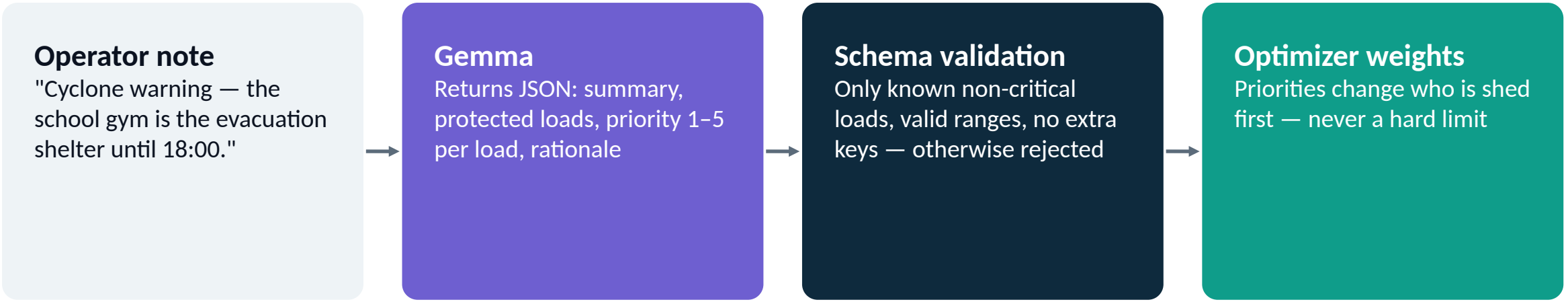
median recovery time (95th percentile 1.2 ms)

22 / 22

re-runs produced identical results

Reproduce: python app/evaluation.py --n 300 --seed 7. Both checkers share the same constraint definitions, so agreement shows the code is consistent — not that the physics model is complete.

Gemma turns human context into validated priorities



Every call is labelled with its real outcome — no model output is ever simulated

inference
real model response, validated

rejected
model answered, output failed validation

unavailable
timeout or HTTP error after 1 retry

not configured
no endpoint set

If Gemma is missing, slow or wrong, the system falls back to declared default priorities. The verifier never reads Gemma's output, so safety never depends on the model.

Feasible today on a small, open Google stack



Gemma
Operator-context understanding (Google AI Studio or self-hosted)



Google OR-Tools
DC optimal power flow — the dispatch plan (GLOP linear solver)



Cloud Run + Secret Manager
Hosting the FastAPI app; Gemma key kept out of the code

≈ 0.2 ms
optimizer solve

≈ 0.1 ms
verifier run

36
automated tests passing

1 vCPU
enough to run the prototype

Not used and not claimed: Google Cloud Pub/Sub (the event stream is in-process) and the NeuroSAN runtime (the agent pipeline is a custom asyncio DAG inspired by it).

From one simulated island to real community grids



Built to scale

- The network is plain data — a new microgrid is a new data file, not new code
- Planning and checking grow polynomially with network size; community grids are small
- One isolated Cloud Run service per community

1. **Real telemetry**
   Pub/Sub or IoT ingestion instead of the simulator
2. **Look-ahead planning**
   multi-hour dispatch so batteries aren't drained too eagerly
3. **Fuller physics**
   voltage and reactive-power checks
4. **Persistent audit trail**
   e.g. Firestore instead of in-memory records
5. **Field validation**
   with a utility partner and a power-systems engineer

What this prototype is — and isn't

It is

- A working simulator with a real optimizer, an independent verifier and replayable audit records
- Deterministic: the same inputs give the same verdicts and hashes
- Tested: 36 automated tests, a 300-scenario randomized evaluation, and live Gemma 4 calls

It isn't (yet)

- Connected to real equipment — all data is synthetic
- Full physics — lossless DC power flow, one 15-minute interval, no voltage or stability
- A formal proof — "certified" means every modeled constraint holds
- Instant at the AI step — a live Gemma call takes several seconds, while planning and checking take under 1 ms

Who benefits



Island & remote communities

Fewer outages of clinics, water and emergency radio during sudden renewable drops — and less diesel burned.



Microgrid operators

A second check that explains exactly why a plan is unsafe before anything is switched.



Anyone adding AI to infrastructure

A working pattern: AI handles messy human context, a deterministic check has the final say.

Gemma reads the context · OR-Tools plans · an independent verifier decides

github.com/souzataniya/SynapseGrid